Exam Date & Time: 16-Nov-2024 (01:15 PM - 04:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

UNIVERSITY THEORY EXAMINATION (REGULAR) NOVEMBER, 2024

SEVENTH SEMESTER OF B.TECH (CE)

Design of Language Processors [CE442]

Marks: 70 **Duration: 195 mins.**

Section I

Answer all the questions. Section Duration: 40 mins

1 Match the components of a compiler to their functions: (2)

1)	Lexical Analyzer		A)	Improves the quality of generated code by reducing redundancy
2)	Syntax Analyzer		B)	Identifies token patterns
3)	Code Generator		C)	Checks the source code for correctness in terms of grammar
4)	Code Optimizer		D)	Translates source code to machine code

Course Outcome: "CO1" / Cognitive Level: "Understand"

2 In compiler terminology, reduction in strength means______. (1)

1)	Replacing a costly operation by a relatively cheaper one	2)	Replacing run time computation by compile time computation	3)	Removing common sub-expressions	4)	Removing loop invariant computation

Course Outcome: "CO6" / Cognitive Level: "Understand"

3 Which of the following is the important factor that justifies theuse of a stack in shift-reduce parsing? (1)

1)	The handle will always appear inside the stack and never on top	2)	The handle will always appear on top of stack and never inside	3)	The handle will never appear in stack	4)	None of the above

Course Outcome: "CO2" / Cognitive Level: "Understand"

4 In a language with nested scopes, how would you modify the symboltable data structure to ensure efficient handling of variablescoping? (1)

1)	Use a global symbol table for all scopes	2)	Use a stack of symbol tables, with each scope having its own table	3)	Use a separate symbol table for each function but merge them during compilation	4)	Use a single linked list with variable scope as an attribute

Course Outcome: "CO3" / Cognitive Level: "Understand"

5 You are implementing a compiler that requires efficient symbollookup for variables in a program. Which of the following datastructures would provide optimal performance in terms of searchtime? (1)

1)	Binary Search Tree	2)	Linked List	3)	Hash Table	4)	Stack

Course Outcome: "CO3" / Cognitive Level: "Analysis"

6 Consider the below grammar (2)
S→ABa | bCA
A→cBCD| ε
B→CdA | ad
C→eC |ε
D→bsf| a
What is follow(B)?

1)	{a,b,d,e}	2)	{a,b,d}	3)	{b,d,e}	4)	{a,b,e}

Course Outcome: "CO2" / Cognitive Level: "Application"

7 Consider the following translation scheme. (2)
S → ER
R → *ER {print ("*");}
R → ε
E → F + E {print ("+");}
E → F
F → (S)
F → id {print(id.value);}
Here id is a token that represents an integer and id.valuerepresents the corresponding integer value. For an input '2 * 3 + 4', thistranslation scheme prints ______.

1)	2 * 3 + 4		2)	2 * +3 4		3)	2 3 * 4 +		4)	2 3 4+*	

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

8 Analysis which determines the meaning of a statement once itsgrammatical structure becomes known is termed as

1)	Semantic analysis		2)	Syntax analysis		3)	Regular analysis		4)	General analysis	

(1)

Course Outcome: "CO1" / Cognitive Level: "Remember"

9 For a C program accessing X[i] [j] [k], the following intermediatecode is generated by a compiler. Assume that the size of an integeris 32 bits and the size of a character is 8 bits.
t0 = i * 1024
t1 = j * 32
t2 = k * 4
t3 = t1 + t0
t4 = t3 + t2
t5 = X[t4]
Which one of the following statements about the source code for theC program is CORRECT? (2)

1)	X is declared as “int X[32] [32] [8]”		2)	X is declared as “int X[4] [1024] [32]”		3)	X is declared as “char X[4] [32] [8]”		4)	X is declared as “char X[32] [16] [2]”	

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

10 In a system using shared libraries, a developer wants to create alibrary that can be loaded at any memory address. How should theyimplement the code to ensure it is position-independent?

1)	Use absolute memory addresses for all function calls		2)	Use relative addressing and indirect jump tables to ensure the code can be loaded at any address		3)	Implement stack based addressing to ensure compatibility across different memory locations		4)	Use global variables for all important data to prevent address conflicts	

(1)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

11 Consider the following **pseudo-code and** convert thegiven **pseudo-code** into **three-address code(TAC)** as an intermediate representation. Identify thecorrect sequence of instructions for one iteration of the innermostloop.

```
for (i = 0; i < n; i++) {
for (j = 0; j < m; j++) {
C[i][j] = 0;
for (k = 0; k < p; k++) {
C[i][j] = C[i][j] + A[i][k] * B[k][j];
}
}
}
```

1)	i = 0 j = 0 C[i][j] = 0 t1 = i * p + k t2 = k * m + j t3 = A[t1] * B[t2] t4 = C[i][j] + t3 C[i][j] = t4		2)	i = 0 j = 0 C[i][j] = 0 t1 = C[i][j] + B[k][j] C[i][j] = t1		3)	i = 0 j = 0 C[i][j] = 0 t1 = i * m + j t2 = A[i][k] * B[k][j] t3 = C[t1] + t2 C[t1] = t3		4)	None of the Above	

(2)

Course Outcome: "CO5" / Cognitive Level: "Remember"

12 Evaluate the performance impact of using dynamic storage allocationin real-time systems. What could be a potential downside?

1)	Increased execution speed		2)	Unpredictable memory fragmentation		3)	Faster access to memory		4)	Less memory overhead	

(1)

Course Outcome: "CO3" / Cognitive Level: "Understand"

13 In given program there are 3 user defined functions (1)
(1) main
(2) f1
(3) f2
In main there is a call of f1 and from f1 there is a call for f2.During program execution, when call for f1 will encounter in mainat that time f1 will be linked and when call of f2 encountered inf1 then f2 will be linked. Choose appropriate from link for givenscenario.

1)	Dynamic linking		2)	Absolute loader		3)	Linkage editor		4)	None of the above	

Course Outcome: "CO3" / Cognitive Level: "Analysis"

14		

Select an appropriate approach for obtaining information about theflow of data along program execution routes.

1)	Dataflow Line		2)	Dataflow Analysis		3)	Both Dataflow Line and Dataflow Analysis		4)	None of the Above	

(1)

Course Outcome: "CO5" / Cognitive Level: "Remember"

15		

Which of the following is not valid three address code?

1)	X = -Y		2)	X = Y		3)	X = Y + Z		4)	X[i] = Y[j]	

(1)

Course Outcome: "CO6" / Cognitive Level: "Understand"

Section II

Answer 5 out of 6 questions.

1		

Represent and analyze the output for the various phases of acompiler with respect to the following assignment statement:

area = length * width + 2 * (length + width)

Assume the variable declarations given as below.
int length, width;
float area;

(5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

2		

Analyze and describe in detail the method utilized in the scanningprocess to reduce processing overhead for each input character. Inyour answer, discuss the techniques or algorithms involved, howthey improve efficiency, and any specific challenges addressed inminimizing this overhead.

(5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

3		

Construct the predictive parsing table for the followinggrammar.
S → aA
A→ bB | ^
B → c | ^
Using this parsing table, demonstrate the steps to parse the inputstring **"**ab**"**.

(5)

Course Outcome: "CO2" / Cognitive Level: "Create"

4		

(a) Compute FIRST (X) & FOLLOW(X) for the all the non-terminals inthe following given grammar.
S → AB | C
A→ a | ε
B→ b
C → c
(b) Eliminate Left Factoring in the following grammar.
S → aAb | aAc
A → d | e

(5)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5		

Construct SLR(1) parsing table for the following grammar:
S → AA
A → aA | b
Provide a detailed explanation of each step involved in buildingthe parsing table, including the computation of First and Followsets, the construction of the canonical collection of LR(0) items,and the final parsing table entries.

(5)

Course Outcome: "CO2" / Cognitive Level: "Create"

6		

Compute the operator precedence matrix and precedence function forthe following grammar.
+, * and id are terminal symbols.
G → E
E → E + T | T
T → T*F | T
F → id

(5)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

Section III

Answer 5 out of 6 questions.

1		

The single pass assembler received the below assembly code asinput. Write the assembler-generated machine code. Show all datastructures' contents that were utilized in this conversionprocedure as well. (5)

```
        START   201
BEG     MOVER   AREG , ='5'
        SUB     AREG , TWO
        MOVEM   AREG , ANS
        LTORG
        MOVER   BREG , ='8'
        ADD     BREG , ='9'
        MULT    BREG , ONE
        MOVEM   BREG , DATA
        STOP
TWO     DC      '2'
ANS     DS      1
ONE     DC      '1'
DATA    DS      1
```

Find OPTAB content below:

OPTAB		
STOP	IS	(00,1)
ADD	IS	(01,1)
SUB	IS	(02,1)
MULT	IS	(03,1)
MOVER	IS	(04,1)
MOVEM	IS	(05,1)
COMP	IS	(06,1)
BC	IS	(07,1)
DIV	IS	(08,1)
READ	IS	(09,1)
PRINT	IS	(10,1)
DS	DL	R#
DC	DL	R#
START	AD	R#
END	AD	R#
ORIGIN	AD	R#
EQU	AD	R#
LTORG	AD	R#

Course Outcome: "CO3" / Cognitive Level: "Understand"

2			Consider the following production rules and semantic rules.	(5)

Production	Semantic Rule
L → E n	Print (E.val)
E → E + T	E.Val = E.val + T.val
E → T	E.val = T.val
T → T * F	T.val = T.val * F.val
T → F	T.val = F.val
F → (E)	F.val = E.val
F → digit	F.val = digit.lexval

Generate annotated parse tree for the input 3 + 8 * 7 +1. Also showdependency graph and write the evaluation order of the same.

Course Outcome: "CO4" / Cognitive Level: "Application"

3 | Analyze the basic block and draw control graph for following codewritten in higher level language.

```
checkSign(x) {
if (x > 0) {
result = "Positive";
} else if (x < 0) {
result = "Negative";
} else {
result = "Zero";
}
return result;
}
```

(5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

4 | Identify the key issues involved in the design of a code generator.Discuss the concepts of reduction in strength and dead codeelimination within the context of code optimization.

(5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

5 | Symbol Table is an important data structure created and maintainedby the compiler. What kind of data is kept in a symbol table? Additionally, describe the data structures that are used toimplement symbol tables along with their benefits and drawbacks.

(5)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

6		

Why there is a need to generate three address code for the belowexpression. Also show different ways of implementation using
(a) Quadruple
(b) Triple and
(c) Indirect Triple.
Expression: b = (a + c) * (c / e) ^ k (5)

Course Outcome: "CO6" / Cognitive Level: "Evaluate"

-----End-----

Event Name	Subject
University Examination Nov -Dec-2024	CE442:Design of Language Processors

#	Course Outcome	Description	Marks
1	CO1	Understand design and processing of different language processor, loaders and linkers	13
2	CO2	Design top-down and bottom-up parsers	18
3	CO3	Identify different memory management schemes of language processors	15
4	CO4	Develop semantic analysis scheme to generate intermediate code	12
5	CO5	Apply different code optimization techniques	10
6	CO6	Develop algorithms to generate code for a target machine	12
		Total	**80**

* NA - CO Not Selected

Event Name	Subject
University Examination Nov -Dec-2024	CE442:Design of Language Processors

#	Level	No of Questions	Marks (%)
1	Remember	3	4(5%)
2	Understand	8	17(21.25%)
3	Application	2	7(8.75%)
4	Analysis	7	23(28.75%)
5	Create	2	10(12.5%)
6	Evaluate	5	19(23.75%)
	Total	27	80 (100%)